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Academic Year: 2021-2022

Programme: B. Tech Computer Science & Business Systems Year: I Semester: II

Subject: Data Structures & Algorithms

Date: 04 July 2022

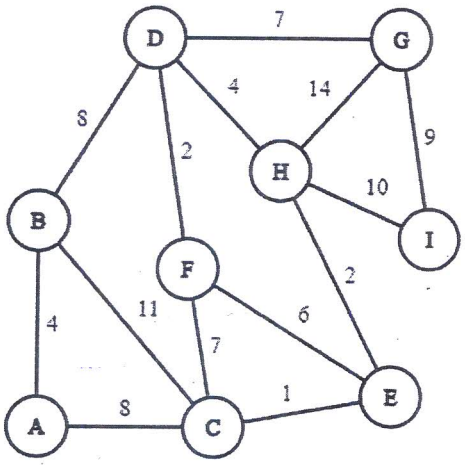
Marks: 100
Time: 11.00 am to 2.00 pm
Durations: 3 (Hrs)
No. of Pages: 3

Re-Examination (2020-21/2021-22)

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

1) Question No. 1 is compulsory.
2) Out of remaining questions, attempt any 4 questions.
3) **In all 5 questions to be attempted.**
4) All questions carry equal marks.
5) **Answer to each new question to be started on a fresh page.**
6) **Figures in brackets on the right hand side indicate full marks.**
7) **Assume Suitable data if necessary.**

Q1 CO- 1; BL-2	a.	Differentiate between linear and non-linear data structures. Identify the most suitable data structure for the given scenario. Provide your justification for the same. i. To store directories and files in a computer. ii. Serving vehicles on toll tax bridge.	[5]
CO - 2; BL- 3	b.	Consider initially empty array-based circular queue of size 5. Show circular queue in each case when the following operations are performed. Also determine front and rear positions at each step a) Add P, Q, R, S, T, U b) Delete two letters c) Add V, W d) Delete four letters and Add X e) ADD Y	[5]
CO -3; BL- 2	c.	Describe the concept of binary threaded tree with an example.	[5]
CO - 4; BL- 3	d.	A hash table of length 10 uses linear probing in case of collision with hash function h(k)=k mod 10. Demonstrate the steps of inserting given values into an empty hash table, the table is as shown below 47, 35, 43, 24, 53, 34	[5]

Q2 CO-3; BL-5	a	Construct the Binary Tree using the following order of traversing: In-order: D B H E I A F J C G Post-order: D H I E B J F G C A i. Perform the pre-order traversal of the constructed tree ii. Find the number of internal nodes in the tree iii. Find the number of nodes in left and right sub tree	[10]
CO-2; BL-4	b	Consider a burger shop outside a park. The shop has only a single window and there is just one person and this person can service only a single customer at a time. The software is made to handle customer request one by one. Write insertion and deletion function of appropriate data structure which their team might have used in the software	[10]
Q3 CO-3; BL-3	a	Below graph represents the connection of cities on a map. Satyam wants to visit all the cities from the source node A. Apply Dijkstra's algorithm to find the shortest path from A to all other vertices. Show all steps  	[10]
CO-1 BL-2	b	i. Define recursion. Identify the data structure used to solve the recursion and explain with an example? ii. Arrange the following algorithm in terms of increasing order of complexity a) Binary search b) Inserting node at the end of linked list c) Addition of matrices	[10]
Q4 CO-2 BL- 4	a	Datamart is purchasing products from different distributors. They are using singly linked list to maintain information of each distributor. a. Write a function to insert a distributor id at the first position b. Write a function to search a distributor by his distributor id	[10]

CO-4; BL-4	b	Perform binary search on the following numbers where the value to be searched is 77. Write program for performing binary search and discuss the complexity of linear and binary search algorithms. 10, 45, 55, 77, 88, 100	[10]
Q5 CO- 2; BL-3	a	Consider the following arithmetic infix expression K + L – M * N + (O ^ P) * W / U / V * T + Q Convert the following expression to postfix expression using stack. Demonstrate by showing all steps.	[10]
CO-3; BL-3	b	Write a program to implement insertion of an element into the binary search tree and performing in order traversal.	[10]
Q6 CO-3; BL-3	a	i. Differentiate between breadth first search and depth first search for the traversing the graph. ii. Discuss various techniques for file organization.	[10]
CO-3; BL-3	b	Perform the selection sort on the following numbers (Show all passes). 78, 2, 67, 8, 1, 2, 90, 8, 81, 7, 6 Write a program to find squared sorted array using selection sort. Example: Input: nums = [-4, -1,0,3,10], Output: [0,1,9,16,100]	[10]